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RESEARCH ARTICLE

Integrating precipitation zoning with random forest regression for the spatial downscaling of satellite-based precipitation: A case study of the Lancang–Mekong River basin

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Downscaling satellite-based precipitation to fine scales is crucial for deepening our understanding of global hydrologic cycles and water-related issues. In this study, a novel approach that integrates precipitation zoning with random forest regression is proposed for the spatial downscaling of satellite-based precipitation. Precipitation zoning is delineated through iterative rotated empirical orthogonal function (REOF) analyses of ground- and satellite-based precipitation observations. Random forest regression is applied to link the satellite-based precipitation to 1-km-resolution predictors such as latitude (Lat), longitude (Lon), elevation, aspect, slope, and the normalized difference vegetation index (NDVI). The accuracy of the resultant downscaled precipitation is evaluated based on five statistical evaluation indices. The performance of the proposed approach is exemplified in the Lancang–Mekong River basin, taking Tropical Rainfall Measuring Mission (TRMM) 3B43 and Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks–Climate Data Record (PERSIANN–CDR) products acquired in 2001 (wet year), 2005 (normal year), and 2009 (dry year) as the databases. The results show that seven precipitation subregions can be roughly distinguished in the study basin. Zoning-based downscaling outperforms non-zoning-based downscaling in terms of accuracy, resulting in statistically significant reductions in root mean square error (RMSE) and mean absolute error (MAE) of 1–17% across the entire basin. Among the selected predictors, the variables Lat and Lon are the most important for precipitation estimation, whereas the remaining variables have lesser and subregion-dependent importance. The proposed approach is promising for generating high-spatial-resolution precipitation data in regions with sparse ground-based observations and differentiated climatic regimes.

KEYWORDS
lancang–mekong river basin, precipitation zoning, random forest regression, REOF analysis, spatial downscaling

1 | INTRODUCTION

Precipitation is a vital element in the Earth system water cycle, which is closely related to meteorological, hydrological, and ecological processes (Harding *et al.,* 2014; Hou *et al.,* 2014). Accurately and efficiently obtaining synoptic-scale precipitation data at higher spatial resolutions is of great significance for the development of precipitation-related models and disaster prevention and reduction against the current background of global warming, extreme weather, and climatic events, which are occurring frequently (IPCC, 2013).

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Recently, satellite-based precipitation products have become an important source of precipitation data and have been widely applied in many domains, such as climate prediction (Spracklen *et al.,* 2012), drought monitoring and forecasting (Tan *et al.,* 2018), and landslide warnings (Brunetti *et al.,* 2018). However, their coarse spatial resolutions of approximately 0.1°–5° are unsuitable for certain research and application needs, such as hydrological simulation and environmental modelling, especially in regions with various climatic regimes and complex terrains.

Accordingly, a wide range of downscaling approaches has been developed in succession to improve the spatial resolution and suitability of precipitation products for various applications. These approaches can be roughly grouped into statistical and dynamical methods (Haylock *et al.,* 2006). Unlike dynamical methods that require the use of regional climate models and require extensive computer time, statistical methods are more prevalent and easy to use because they only employ a variety of regression models (Foody, 2003; Immerzeel *et al.,* 2009; Jia *et al.,* 2011; Verlinde, 2011; Chen *et al.,* 2014; Shi *et al.,* 2015; Ochoa *et al.,* 2016; Zhao *et al.,* 2018) to establish the statistical relationship between precipitation at a coarse scale and geospatial variables at a finer resolution to estimate precipitation at finer scales (Atkinson, 2013; Hunink *et al.,* 2014). Previous studies have shown that statistical downscaling has good performance in improving the spatial resolution and suitability of satellite-based precipitation. In particular, the random forest (RF) regression algorithm has been demonstrated to be superior to other traditional methods for statistical downscaling of precipitation in recent works (Shi *et al.,* 2015; Jing *et al.,* 2016). Nevertheless, the statistical relationship between precipitation and geospatial variables is region-dependent and varies by case study. For example, precipitation is dominated by elevation in the Qaidam basin in China (Jia *et al.,* 2011) but is less affected by elevation in the Pangani basin in Tanzania (Verlinde, 2011). In addition, large intra-area spatial differences in the relationship between precipitation and geospatial variables exist across vast regions. For instance, the normalized difference vegetation index (NDVI)–precipitation relationship is inconsistent when comparing the entire semi-arid region of southern Botswana and its four subareas due to diverse soil conditions (Grist *et al.,* 1997). Similar results have also been reported by Foody (2003) and Chen *et al*. (2014). Foody (2003) observed that a strong and statistically significant NDVI–precipitation relationship occurred for the entire region of North Africa and the Middle East; however, this relationship does not exist in local parts of the region. Chen *et al.* (2014) found that NDVI–precipitation relationships between the entire region of North China and its individual subregions are variable. Therefore, no single relationship is suitable for all regions or all parts of a vast region.

To cope with the spatial inconsistency of the relationship between precipitation and geospatial variables, we propose a zoning-based downscaling approach by integrating precipitation zoning based on an iterative rotated empirical orthogonal function (REOF) analysis with random forest regression in this study. The objective of this study is to demonstrate the performance of this approach by using the Lancang–Mekong River basin as the study site. To achieve this goal, we first conduct precipitation zoning using the REOF analysis via ground- and satellite-based precipitation observations. Then, the performances of the zoning- and non-zoning-based downscaling approaches are evaluated and compared based on validation statistical indices, including the root mean square error (RMSE), bias (BIAS), coefficient of determination (R^2), mean absolute error (MAE), and mean relative error (MRE). Finally, the relative importance of the geospatial variables used for precipitation estimation is examined for individual precipitation subregions and the entire river basin.

2 | MATERIALS AND METHODS

2.1 | Study area

The Lancang–Mekong River basin (Figure 1) is a typical north–south-oriented basin that has a total area of approximately 795,000 km^2, which covers parts of Southwest China, Myanmar, Laos, Cambodia, Thailand, and Vietnam (MRC, 2005). It spans six climate zones, from cold temperate zones to tropical zones, and expands across diverse topographic regions, including plateaus, valleys, mountains, lowlands, estuarine deltas and others, from north to south (Gupta, 2009). Accordingly, the major land use and cover types consist of grasslands, mixed forests, evergreen/deciduous and needle/broad-leaf forests, woody lianas, and croplands, which dominate these units (Rundel, 2009). Because of its unique geographical situation and complex terrain, the entire basin is dominated by the East Asian and Indian monsoons during the wet season (May to October) but is mainly controlled by mid-latitude westerlies in the dry season (November to April) (Fan and He, 2015). Precipitation in this basin is strongly affected by monsoons, terrain, topography, and other underlying surface factors, such as vegetation cover (Zhang *et al.,* 2015). The annual precipitation ranges from approximately 300 to 5,000 mm from the plateaus in the north to the delta in the south (Rundel, 2009; Räsänen, 2014).

2.2 | Data set

In this study, two satellite-based precipitation products (i.e., the Tropical Rainfall Measuring Mission (TRMM) 3B43 V7 precipitation product (hereafter referred to as TRMM) and the Precipitation Estimation from Remotely Sensed Information using Artificial Neural Networks–

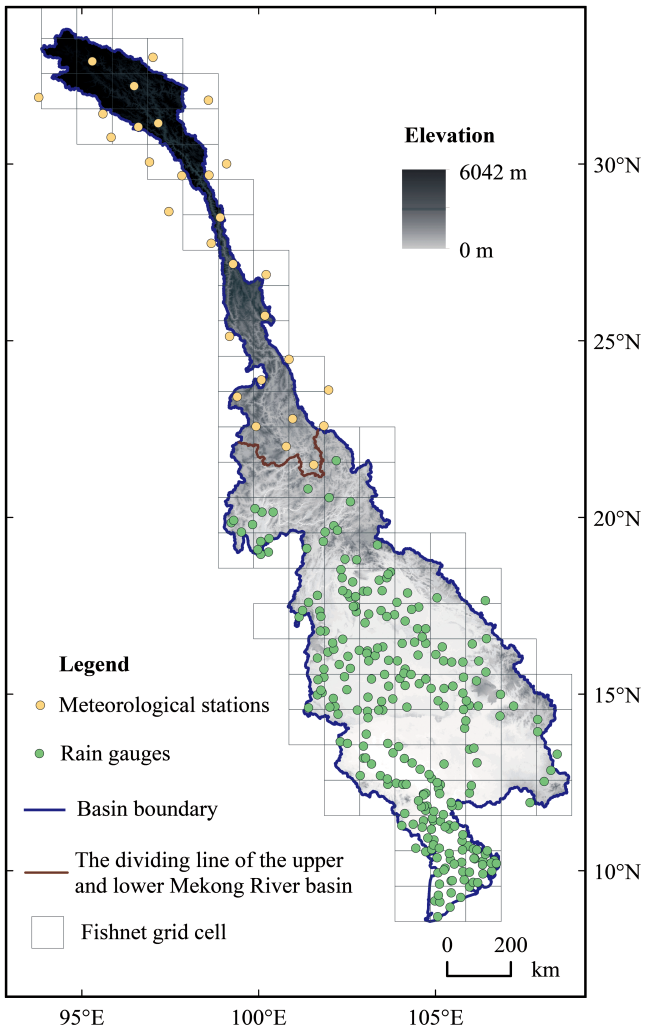

FIGURE 1 Spatial distributions of meteorological stations and rain gauges in the Lancang–Mekong River basin [Colour figure can be viewed at wileyonlinelibrary.com]

Climate Data Record (PERSIANN–CDR) precipitation product (hereafter referred to as CDR)), the Moderate Resolution Imaging Spectroradiometer (MODIS) MOD13A3 NDVI product, satellite-based digital elevation model (DEM) data, and ground-based precipitation data were used.

TRMM and CDR have a spatial resolution of 0.25° and a temporal resolution of 1 month (Ashouri *et al.,* 2015; MacRitchie *et al.,* 2017) and are downloadable from the websites http://mirador.gsfc.nasa.gov and ftp://data.ncdc.noaa.gov, respectively. These data show good accuracy at both spatial and temporal scales across the Lancang–Mekong River basin (Zeng *et al.,* 2012a, 2012b; He *et al.,* 2017; Wang *et al.,* 2017; Chen *et al.,* 2018). The monthly TRMM and CDR data from January 2000 to December 2014 were used in this study. The annual precipitation is a summation of the monthly precipitation in a year.

The DEM data, with a spatial resolution of 1 km, were obtained from the United States Geological Survey (USGS) website (https://lta.cr.usgs.gov/HYDRO1KReadMe). The latitude (Lat), longitude (Lon), elevation, slope, and aspect were all extracted from the DEM. These extracted data were then resampled to 0.25° individually.

The MODIS MOD13A3 1-km-resolution NDVI data from January 2001 to December 2014, which are downloadable from https://modis.gsfc.nasa.gov/data/, were used in this study. This product was synthesized by the original MODIS NDVI at 30 days for the current month (Didan *et al.,* 2015). The mean annual NDVI was generated by averaging the monthly synthetic NDVI. The mean annual NDVI was resampled to 0.25°.

Ground precipitation observational data were collected from 29 meteorological stations in the upper Mekong basin and its surroundings and 261 rain gauge stations in the lower Mekong basin and its surroundings (Figure 1). Monthly precipitation data recorded at the meteorological stations from 2000 to 2014 were sourced from the Chinese Meteorological Science Data Sharing Service Platform (http://data.cma.cn/). Daily precipitation data recorded in rain gauges from 2001 to 2005 and in 2009 were sourced from the work of Wang *et al.* (2016). Only 195 of these rain gauges had continuous records from 2001 to 2005. The annual precipitation data were obtained from a summation of monthly or daily precipitation in a year.

2.3 | Methods

2.3.1 | REOF analysis

An REOF analysis is also known as a rotated principal component analysis (RPCA) (Richman, 1986). This method and its derivatives are widely used in many fields, such as meteorology and climatology (White *et al.,* 1991; Ujeneza and Abiodun, 2015). The varimax REOF analysis proposed by Kaiser (1958) was used to identify the predominant regional patterns of precipitation variabilities across the Lancang–Mekong River basin in this study. This REOF analysis maximizes the variance of the squared correlation between each rotated principal component and variable to provide the simplest pattern description while explaining the maximum amount of variance. Detailed descriptions of the varimax REOF analysis can be found in several reference papers (Kaiser, 1958; Richman, 1986; White *et al.,* 1991; Ujeneza and Abiodun, 2015).

REOF analyses postulate that the employed data series is homogeneous in space and time domains and that the number of samples is larger than the number of stations (Karl *et al.,* 1982; Conroy and Overpeck, 2011). Considering the uneven spatial distribution and poor continuity (i.e., shorter than 60 months) of ground precipitation observations in the lower Mekong basin, it is very difficult to correctly define precipitation subregions at a basin-wide scale based only on an REOF analysis of ground-observed precipitation. Therefore, an iterative REOF analysis was separately performed on ground- and satellite-based precipitation using monthly precipitation time series data. The REOF analysis of ground-based precipitation was conducted individually in the upper

and lower regions of the Lancang–Mekong River basin, and the REOF analysis of satellite-based precipitation was performed for the entire basin. For each data source, the varimax REOF analysis was iteratively performed until only one dominant empirical orthogonal function (EOF) mode with an eigenvalue >1.0, explaining more than 75% of the total variance, occurred in each precipitation subregion. The iterative REOF analysis of ground- and satellite-based precipitation aims to define more homogeneous precipitation subregions. To make the ground- and satellite-based precipitation as comparable as possible, we divided the entire basin into 114 grid boxes of 1° × 1° (c.a. 111 km × 111 km), which is approximately the average ground-based observational station spacing in the upper regions of the Lancang–Mekong River basin (Figure 1). Only 75 of these grid boxes have ground-based precipitation stations. The value of gridded precipitation for the centre of each grid box was computed by averaging precipitation values over the ground-based stations or the satellite-based pixels within it.

The final boundaries of precipitation subregions were generated through blending the zoning results of the three data sources. Due to its relatively high reliability, ground-based precipitation data were used to delimit the initial boundaries of precipitation subregions through the iterative REOF analysis. Then, the boundaries of those precipitation subregions with sparse ground-based observations were amended with the subregion boundaries generated from the iterative REOF analysis of TRMM data, which can better capture precipitation extremes across the Lancang–Mekong River basin (Chen *et al.,* 2018). Finally, the boundaries of precipitation subregions were further adjusted by reference to the boundaries of topographic watershed divides. The divides of higher order basins were given higher priority for delineating the final boundaries.

2.3.2 | RF regression model

RF is an ensemble learning algorithm that has been widely used for regression analyses (Breiman, 2001; Liaw, 2012; Shi *et al.,* 2015; Jing *et al.,* 2016). A random forest is a combination of a large number of trees. Each tree within a random forest is independently constructed using a bootstrap sample of the original data set (Figure 2a), and each node is split using the best random selection of predictor variables at that node (Breiman, 2001; Liaw and Wiener, 2002). The number of trees (*ntree*) in the forest and the number of predictor variables in the random subset at each node (*mtry*) are the only two user-defined parameters required for an RF model (Figure 2a and b). The resultant regression prediction of a forest is equal to the average of the predictions from the *ntree* trees. Overall, a suitably large *ntree* can produce a more stable and robust prediction (Breiman, 2001; Liaw and Wiener, 2002). The importance of the predictor variables is a key by-product of an RF model; it can be defined as the mean decrease in the prediction error for out-of-bag samples (i.e., the observations left out of the bootstrap samples) after permutating each predictor variable or the total decrease in node impurities (i.e., the variance of all data points that are routed through that node) as measured by the averaged residual sums of squares across all trees (Liaw, 2012).

The performance of an RF regression algorithm is schematically expressed as follows (Breiman, 2001; Liaw and Wiener, 2002): (a) select *ntree* samples from the original data using the bootstrap method; (b) use each of the *ntree* bootstrap samples to grow a non-pruning regression tree, and at each node, choose the best split from among randomly sampled *mtry* of the predictor variables (Figure 2b); (c) generate a new prediction by averaging the predictions of the *ntree* trees. The abovementioned procedure is also illustrated in Figure 2.

In the present study, Lat, Lon, elevation, aspect, slope, and the NDVI are input as predictor variables, and satellite-based precipitation is used as the dependent variable. The regression model was trained and validated over the 0.25°-resolution data and applied to the 1-km-resolution data to predict precipitation at the 1-km-resolution scale. Regarding the zoning-based approach (Zrf), the regression model was individually built for each precipitation subregion. The value of *ntree* was set as 1,000, and *mtry* was equal to the square root of the number of predictor variables. A permutation test was used to assess the importance measures of the predictor variables, which describe the increase in mean squared error (MSE) (% IncMSE). The RF regression was conducted using an RF package in R provided by Liaw (2012). Detailed information on the procedure of the RF regression based on the randomForest package in R can be found in previous studies (Liaw, 2012; Shi *et al.,* 2015; Jing *et al.,* 2016). Because adding the residuals between the original and fitted precipitation values to the RF regression model rarely improves the accuracy of the precipitation estimation, as suggested by Shi *et al.* (2015) and Jing *et al.* (2016), the fitted residuals were not considered in this study.

2.3.3 | Validation with the original products and ground observations

Several popular statistical indices, including the RMSE, BIAS, R^2, MAE, and MRE, are used to quantitatively evaluate the performance of the Zrf against that of the non-zoning-based approach (Drf). The Zrf is applied to each individual precipitation subregion, whereas the Drf takes the whole river basin as one region. The calculation formulas for these statistical indices are defined as follows:

$$\mathrm{RMSE}=\sqrt{\sum_{i=1}^{n}\frac{(D_i-P_i)^2}{n}} \tag{1}$$

$$\mathrm{BIAS}=\frac{\sum_{i=1}^{n}D_i}{\sum_{i=1}^{n}P_i}-1 \tag{2}$$

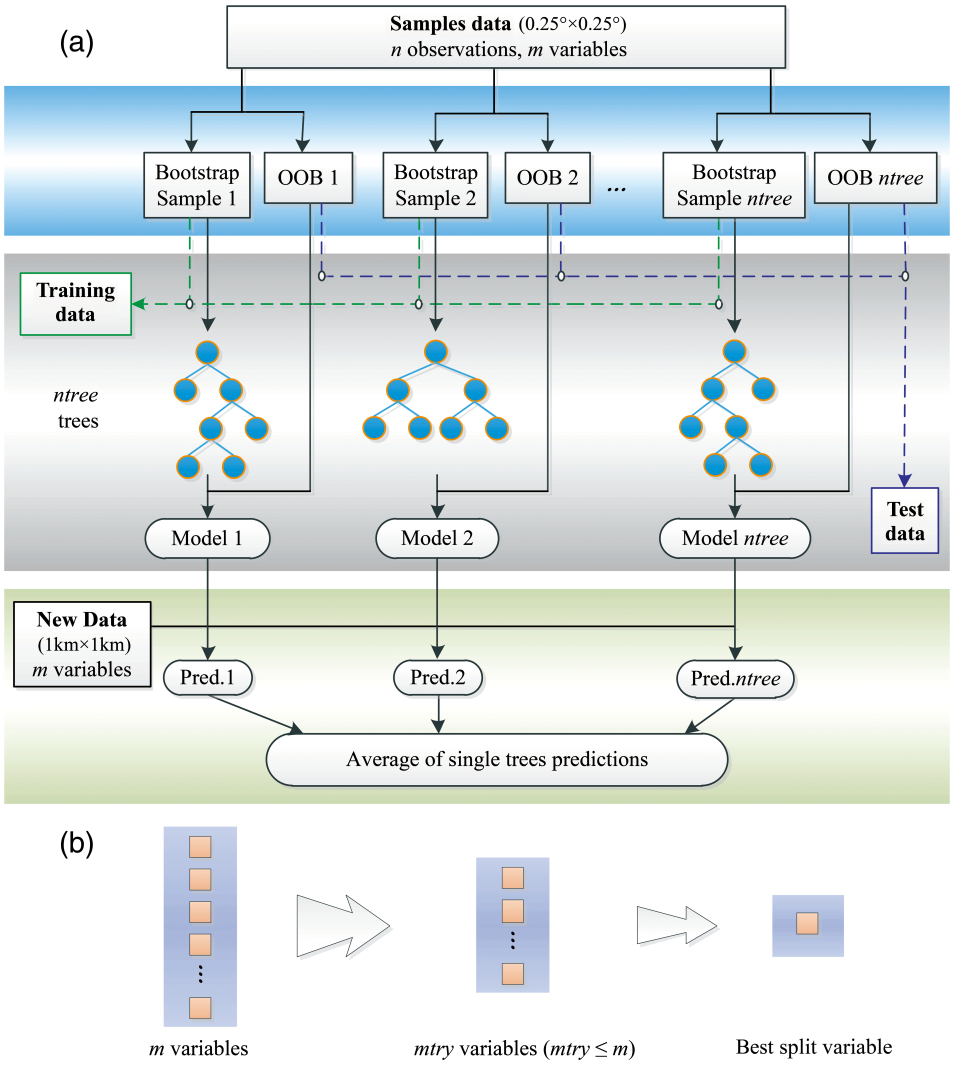

FIGURE 2 Flowchart of random forest (RF) for regression (a) and the selection process of the best split at each node (b) [Colour figure can be viewed at wileyonlinelibrary.com]

$$R^2 = \left[\frac{\sum_{i=1}^{n}(P_i - \overline{P}) \sum_{i=1}^{n}(D_i - \overline{D})}{\sqrt{\sum_{i=1}^{n}(P_i - \overline{P})^2} \sqrt{\sum_{i=1}^{n}(D_i - \overline{D})^2}} \right]^2 \quad (3)$$

$$\mathrm{MAE} = \frac{\sum_{i=1}^{n} |D_i - P_i|}{n} \quad (4)$$

$$\mathrm{MRE} = \frac{\sum_{i=1}^{n} \frac{|D_i - P_i|}{P_i}}{n} \quad (5)$$

where P_i denotes the original ground- or satellite-based precipitation in the i-th pixel at 1 km × 1 km grid scale, D_i denotes the downscaled precipitation in the i-th pixel at 1 km × 1 km grid scale, and n denotes the total number of pixels.

The relative differences (Δ) in RMSE and MAE are calculated with the following formulas:

$$\Delta_{\mathrm{RMSE}} = \frac{\mathrm{RMSE_{Zrf}} - \mathrm{RMSE_{Drf}}}{\mathrm{RMSE_{Drf}}} \times 100\% \quad (6)$$

$$\Delta_{\mathrm{MAE}} = \frac{\mathrm{MAE_{Zrf}} - \mathrm{MAE_{Drf}}}{\mathrm{MAE_{Drf}}} \times 100\% \quad (7)$$

where the subscripts Drf and Zrf denote non-zoning- and zoning-based approach, respectively. Subsequently, the statistical significances of these differences between zoning- and non-zoning-based downscaling were assessed with a paired Wilcoxon Signed-Rank (WSR) test (at the significance level of 0.05) via the R package (Torsten *et al.*, 2017).

Validations of downscaled precipitation against original satellite-based precipitation and ground-based observational precipitation were conducted for the representative years of 2001 (wet year), 2005 (normal year) and 2009 (dry year). Only the meteorological stations and rain gauges within the boundary of the Lancang–Mekong River basin were considered as candidate validation sites. As a result, precipitation data recorded at

12 meteorological stations and 229 rain gauges in 2001, 200 rain gauges in 2005, and 24 rain gauges in 2009 were used for validation.

3 | RESULTS

3.1 | Precipitation zoning

As shown in Figure 3, six to seven precipitation subregions were delimited separately based on ground-based precipitation (Figure 3a) and TRMM (Figure 3b) and CDR (Figure 3c) satellite-based precipitation. Four of these subregions were roughly and consistently delineated from these three precipitation data sources. They were located in the upper (G–I, T–I, and C–I) and middle (G–II, T–II, and C–II) regions of the Lancang River basin, the lower regions of the Lancang River basin and the upper regions of the Lower Mekong River basin (G–III, T–III, and C–III), and Mekong Delta (G–VI, T–VII, and C–VII). However, inconsistent precipitation subregions occurred in the lower Mekong regions between the mountains of the northern Lao People's Democratic Republic (PDR) and Thailand and in the Mekong Lowlands.

Based on the spatial integration of precipitation subregions generated from these three precipitation data sources, a total of seven precipitation subregions were defined for the entire Lancang–Mekong River basin (Figure 3d). The above-mentioned consistent subregions were retained and partially adjusted to the upper (I), middle (II), and lower (III) Lancang River basin subregions and Mekong Delta subregion (VII). The inconsistent regions were further divided into three precipitation subregions: a subregion containing the mountains of the northern Lao PDR, Thailand and the Annamite Range (IV); the Khorat Upland subregion (V); and the Mekong Lowland subregion (VI). The between-subregion differences in precipitation were overall statistically significant at the 95% confidence level, although the within-subregion

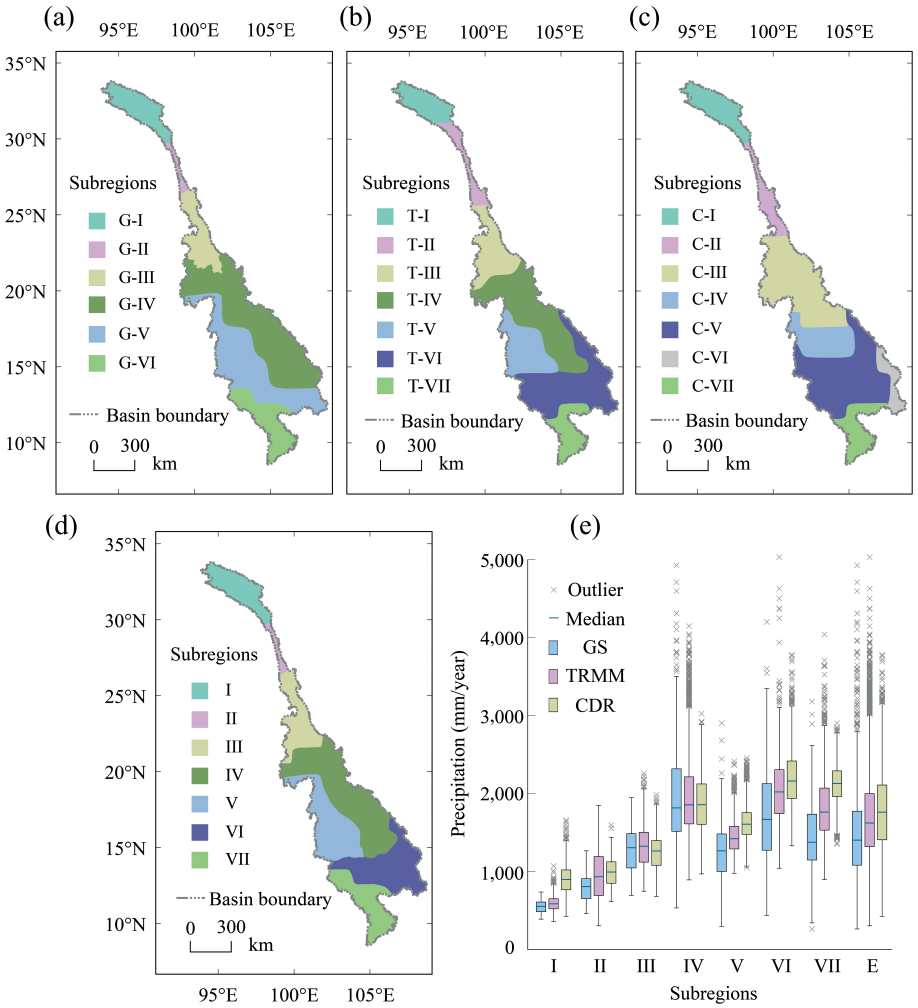

FIGURE 3 Precipitation zoning based on precipitation data from ground observations (GS) (a), Tropical Rainfall Measuring Mission (TRMM) (b) and Climate Data Record (CDR) data (c), the resultant precipitation zoning (d), and the statistical characteristics of ground- and satellite-based (TRMM and CDR) precipitation (e) for individual precipitation subregions (I, II, III, IV, V, VI, and VII) and the entire basin (E) in the Lancang–Mekong River basin. The uppercase letters and Roman numerals that make up the names of the precipitation subregions in (a)–(c) denote the precipitation data type and precipitation subregion, respectively. The boxplot in (e) describes the median (the central horizontal line), the 25th and 75th percentiles (the edges of the box), the most extreme data points (the whiskers) and the outliers (the product signs) [Colour figure can be viewed at wileyonlinelibrary.com]

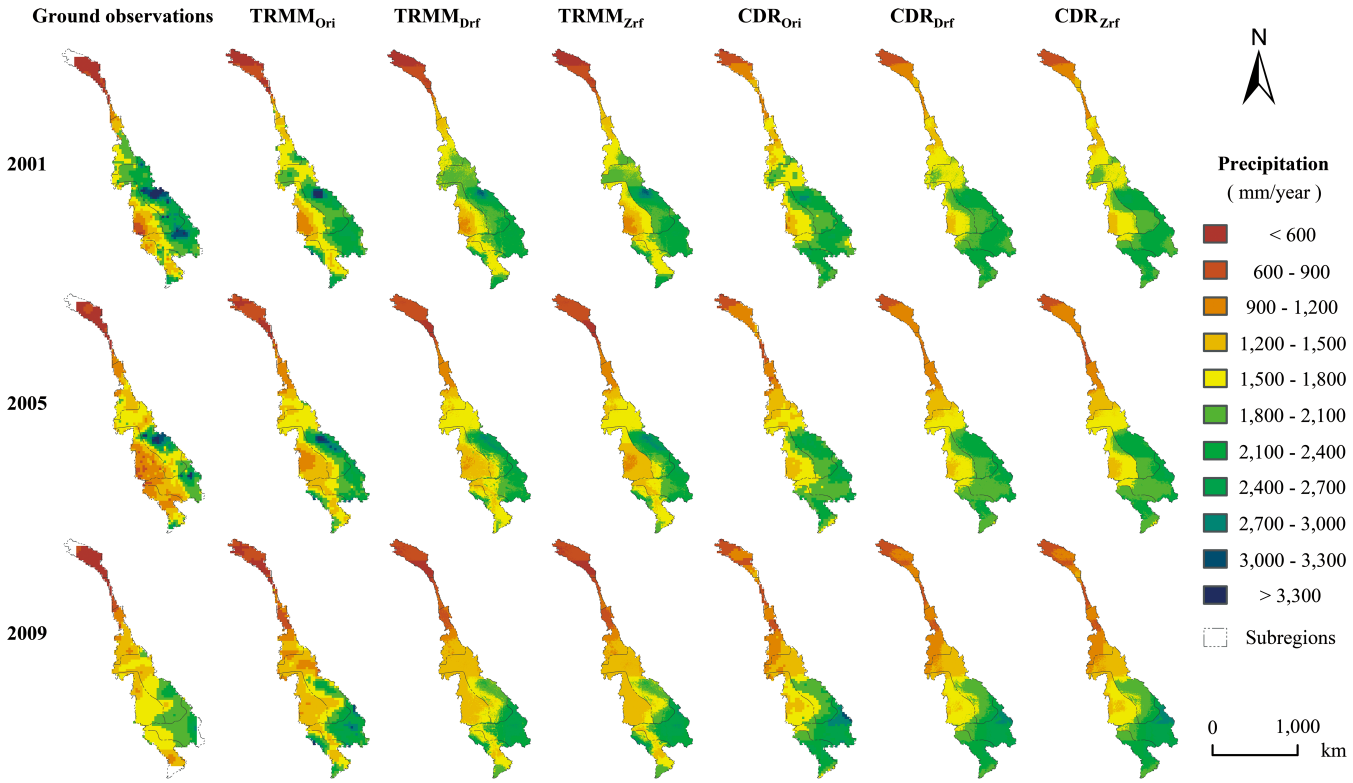

FIGURE 4 The downscaled precipitation at a spatial resolution of 1 km across the Lancang–Mekong River basin in 2001, 2005, and 2009. The subscripts Ori, Drf, and Zrf denote the original 0.25°-resolution precipitation, and the non-zoning- and zoning-based downscaled 1-km-resolution precipitation, respectively. The individual annual precipitation maps based on ground observations are generated by the inverse distance weighting (IDW) interpolation algorithm [Colour figure can be viewed at wileyonlinelibrary.com]

dissimilarities in precipitation from the three data sources were also considerable in the subregions from IV to VII, where ground-based observations and the TRMM product have higher interannual precipitation variations than the CDR product (Figure 3e).

3.2 | Precipitation downscaling and its validation

3.2.1 | Downscaled results

As shown in Figure 4, the spatial patterns of annual precipitation obtained from the ground observations and the original and downscaled satellite-based TRMM and CDR data over the entire Lancang–Mekong River basin were roughly similar. The amount of annual observational precipitation generally decreased from 299 to 4,925 mm along the latitudinal gradient, and more intense precipitation (2700–4,925 mm) mainly occurred over the southeastern parts of the basin. The differences in the amount of precipitation among the ground-based and original and downscaled satellite-based precipitation maps were mainly observed in subregions IV, VI, and VII. Compared to the ground-based precipitation map, the satellite-based precipitation maps had smaller regions with annual precipitation greater than 2,700 mm (subregions IV and VI) and regions with annual precipitation less than 1,800 mm (subregion VII), especially for the CDR precipitation maps. It was also found that precipitation downscaling led to a substantial decrease in areas that had an original annual precipitation value less than 900 mm or greater than 2,700 mm via the TRMM and CDR products, respectively. In comparison, the precipitation maps generated by the zoning-based downscaling approach matched better with the original maps than those generated via the Drf.

3.2.2 | Validation with the original products

The scatter plots in Figure 5 show that the amounts of precipitation obtained by the zoning-based downscaling approach had a higher correlation with the original TRMM and CDR precipitation amounts than those via the non-zoning-based downscaling approach. Compared to the downscaled TRMM data, the downscaled CDR precipitation data fits better with the original values. The zoning-based downscaling approach produces much smaller precipitation underestimations, which mainly occur in grids with annual precipitation values greater than 3,000 mm via the TRMM precipitation product. In addition, smaller precipitation underestimations are also observed in 2009 for both satellite-based precipitation products.

As shown in Table 1, the statistical evaluation indices of RMSE and MAE are lower for zoning-based downscaled precipitation than for non-zoning-based downscaled precipitation. The RMSE and MAE are significantly (at the

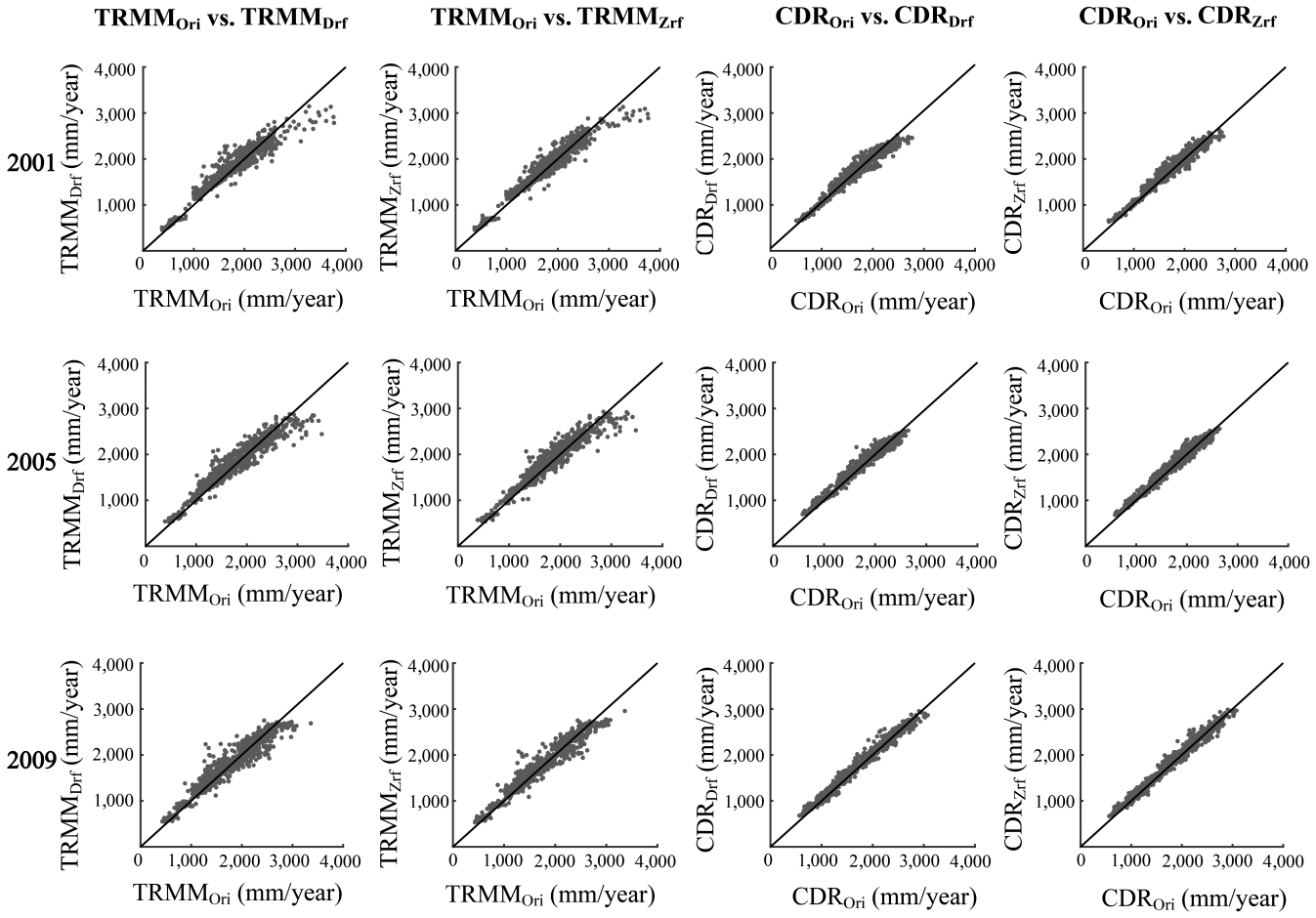

FIGURE 5 Scatter plots of the Tropical Rainfall Measuring Mission (TRMM) and Climate Data Record (CDR) original precipitation products versus their non-zoning- and zoning-based downscaled precipitation results across the entire basin in 2001, 2005, and 2009. The original precipitation products are on the x-axis, and downscaled precipitation is on the y-axis

TABLE 1 Validation accuracies of the non-zoning- and zoning-based downscaled Tropical Rainfall Measuring Mission (TRMM) and Climate Data Record (CDR) precipitation results against the original products across the entire basin in 2001, 2005, and 2009

		RMSE (mm/year)			BIAS		MAE (mm/year)			MRE		R^2	
Product type	**Climate year**	**Drf**	**Zrf**	**Δ(%)**	**Drf**	**Zrf**	**Drf**	**Zrf**	**Δ(%)**	**Drf**	**Zrf**	**Drf**	**Zrf**
TRMM	2001	154	132	−14	0.00	0.00	103	89	−13	0.06	0.05	0.93	0.95
	2005	153	135	−12	0.00	0.00	103	89	−14	0.06	0.05	0.93	0.95
	2009	150	132	−12	0.00	0.00	103	91	−11	0.07	0.06	0.94	0.95
CDR	2001	102	85	−17	0.00	0.00	76	63	−17	0.04	0.04	0.95	0.96
	2005	84	73	−14	0.00	0.00	62	54	−13	0.04	0.03	0.97	0.97
	2009	80	76	−5	0.00	0.00	59	55	−6	0.04	0.03	0.98	0.98

Note. MAE: mean absolute error; RMSE: root mean square error.

significance level of 0.05) reduced by 5 to 17% using the zoning-based downscaling approach. The downscaled CDR precipitation has weaker statistical evaluation indices (RMSE, MAE, and MRE) than those of the downscaled TRMM precipitation. Among the three examined representative years, a more marked reduction in RMSE and MAE occurred in 2001 and 2005, with negative relative differences of 12–17% and 13–17%, respectively.

Figure 6 further shows how the statistical evaluation indices of RMSE and MAE vary over the seven precipitation subregions. However, the behaviours of the downscaled TRMM and CDR precipitation data sets are different. Both the downscaled TRMM and CDR precipitation data sets have the lowest RMSE and MAE values in subregion I. Subregions III and V have smaller RMSE and MAE values than those in subregions II, IV, VI, and VII via the TRMM product, whereas there are no obvious differences in the RMSE and MAE values among these subregions via the downscaled CDR precipitation data set. In addition, the downscaled CDR precipitation in subregions II, IV, VI, and VII generally provides smaller RMSE and MAE values than the downscaled TRMM precipitation.

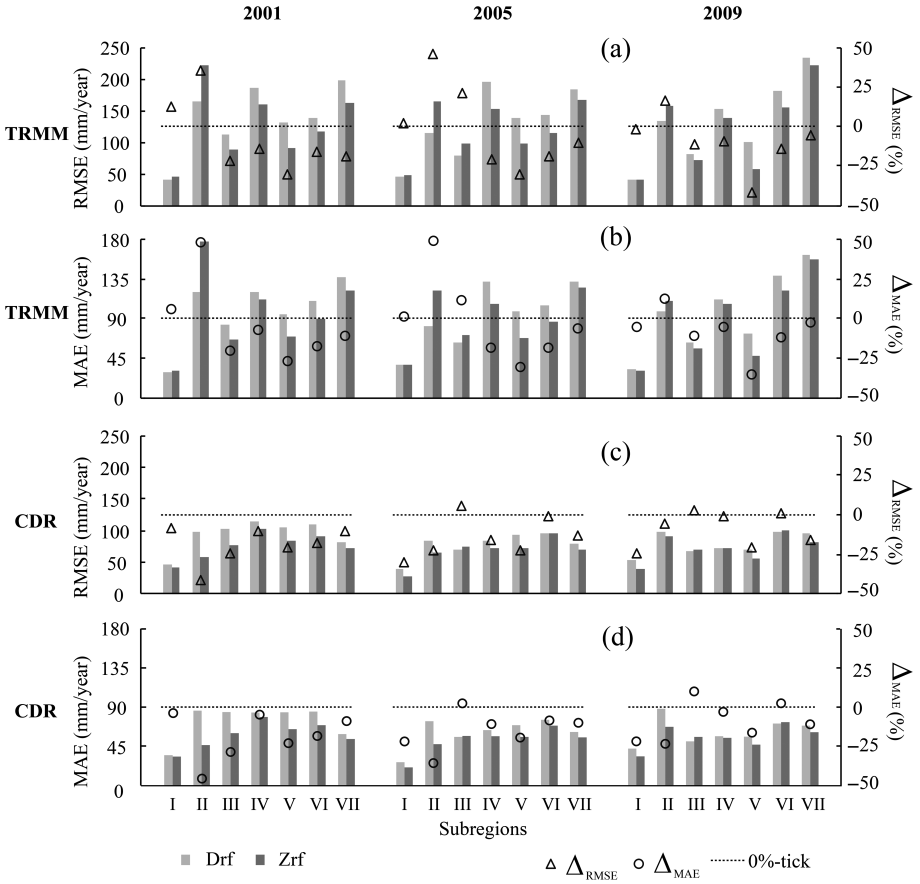

FIGURE 6 The absolute values (left y-axis) and relative differences (right y-axis) of root mean square error (RMSE) (a, c) and mean absolute error (MAE) (b, d) for the downscaled non-zoning- and zoning-based Tropical Rainfall Measuring Mission (TRMM) and Climate Data Record (CDR) precipitation results in all of the individual precipitation subregions (I, II, III, IV, V, VI, and VII) in 2001, 2005, and 2009

The zoning-based downscaling approach results in various reductions in RMSE and MAE over most of the precipitation subregions (Figure 6), with an exception of subregion II for all of the investigated years, subregion I in 2001, and subregions I and III in 2005 for the downscaled TRMM and subregion III in 2005 and 2009 and subregion VI in 2009 for the downscaled CDR data. The relative differences in the RMSE and MAE fluctuated greatly over different subregions and among different years. Compared to the downscaled TRMM precipitation, the downscaled CDR precipitation produced larger reductions in RMSE and MAE in subregions I and II, with relative differences from −4 to −46%.

3.2.3 | Validation with ground observations

Figure 7 shows that overestimations in precipitation occur at the ground-based stations, with precipitation values below 1,000 mm, and underestimations in precipitation at those stations also occur, with precipitation values above 2,500 mm. The zoning-based downscaling approach produces slightly smaller deviations in precipitation, especially via the TRMM precipitation data.

As shown in Table 2, the zoning-based downscaling approach has a slight impact on the statistical evaluation indices via the ground-based observations. Among these evaluation indices, RMSE and MAE are significantly (at the significance level of 0.05) reduced by approximately 1 to 5%. Compared to the downscaled CDR precipitation, the downscaled TRMM precipitation has a higher coefficient of determination (R^2) and smaller values in terms of the other statistical evaluation indices.

3.3 | Relative importance of precipitation predictors

As shown in Figure 8, the relative importance of the selected precipitation predictor variables varies among precipitation subregions and at different study scales. Overall, the variables of Lat and Lon have higher importance, and the aspect variable is unimportant over the entire basin and for the seven precipitation subregions. At the entire basin scale, NDVI has greater importance than elevation. However, the opposite is true in the precipitation subregions, excluding subregions I and IV. In addition, the relative importance of all of the predictor variables is less than 25% in subregion II.

4 | DISCUSSIONS

4.1 | Precipitation zoning based on the iterative REOF analysis of ground- and satellite-based precipitation

The characteristics and dynamics of spatial differentiation are the focus of many natural science disciplines, especially

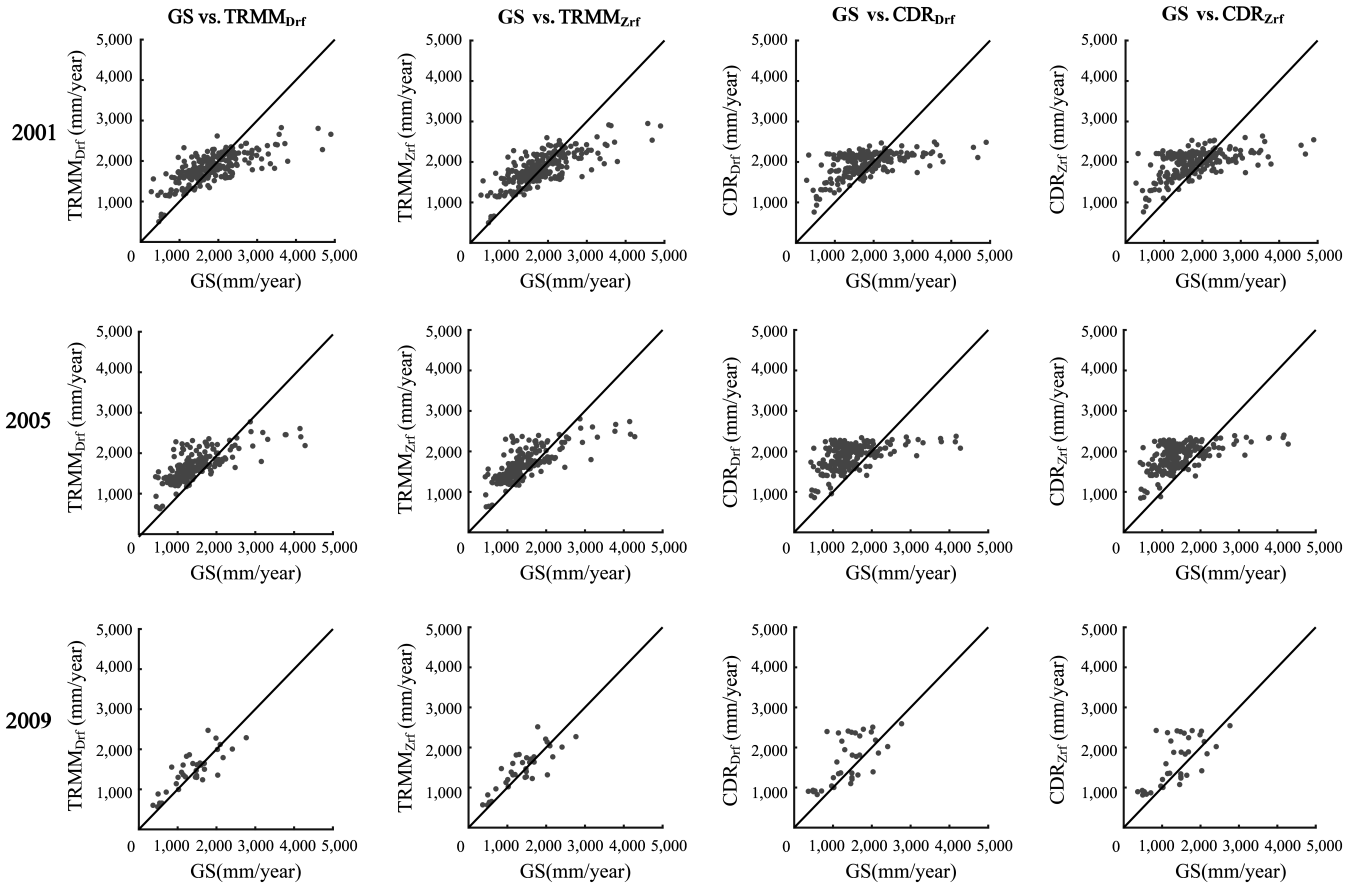

FIGURE 7 Scatter plots of ground-observed precipitation (GS) versus non-zoning- and zoning-based downscaled Tropical Rainfall Measuring Mission (TRMM) and Climate Data Record (CDR) precipitation across the entire basin in 2001, 2005, and 2009. Ground-observed precipitation is on the x-axis, and the downscaled precipitation is on the y-axis

TABLE 2 Validation accuracies of the non-zoning- and zoning-based downscaled Tropical Rainfall Measuring Mission (TRMM) and Climate Data Record (CDR) precipitation results against the ground-observed precipitation results across the entire basin in 2001, 2005, and 2009

		RMSE (mm/year)			BIAS		MAE (mm/year)			MRE		R^2	
Product type	**Climate year**	**Drf**	**Zrf**	**Δ(%)**	**Drf**	**Zrf**	**Drf**	**Zrf**	**Δ(%)**	**Drf**	**Zrf**	**Drf**	**Zrf**
TRMM	2001	550	521	−5	−0.02	−0.02	397	375	−5	0.25	0.24	0.52	0.56
	2005	494	478	−3	0.10	0.10	364	351	−4	0.29	0.28	0.51	0.53
	2009	319	315	−1	0.02	0.02	250	241	−3	0.20	0.18	0.68	0.69
CDR	2001	659	653	−1	0.07	0.07	509	503	−1	0.36	0.35	0.27	0.28
	2005	677	665	−2	0.24	0.24	564	554	−2	0.47	0.46	0.23	0.27
	2009	538	532	−1	0.17	0.16	427	418	−2	0.37	0.36	0.42	0.42

Note. MAE: mean absolute error; RMSE: root mean square error.

geography, meteorology, and climatology (White *et al.,* 1991; Hattab *et al.,* 2015; Ujeneza and Abiodun, 2015; Dos Santos *et al.,* 2016; Kurina *et al.,* 2018). Zoning based on an REOF analysis is an important way to understand the mechanisms of spatial differentiation (White *et al.,* 1991; Ujeneza and Abiodun, 2015). The entire Lancang–Mekong River basin is divided into seven precipitation subregions according to the results of the REOF analysis for ground- and satellite-based precipitation. The resultant zoning is in sound agreement with the physiographic zoning of the Lancang–Mekong River basin (Leinenkugel *et al.,* 2013), which is generated from the multi-source data set that includes Shuttle Radar Topography Mission (SRTM) topographic elevation, annual long-term precipitation, and proportional land cover. It also suggests that the regimes for precipitation in the Lancang–Mekong River basin are strongly constrained by physiographic factors.

Precipitation subregions generated from the three precipitation data sources are inconsistent over the lower Lancang–Mekong River basin. Between approximately 12° and 20°N, in particular, ground-based observations and the TRMM

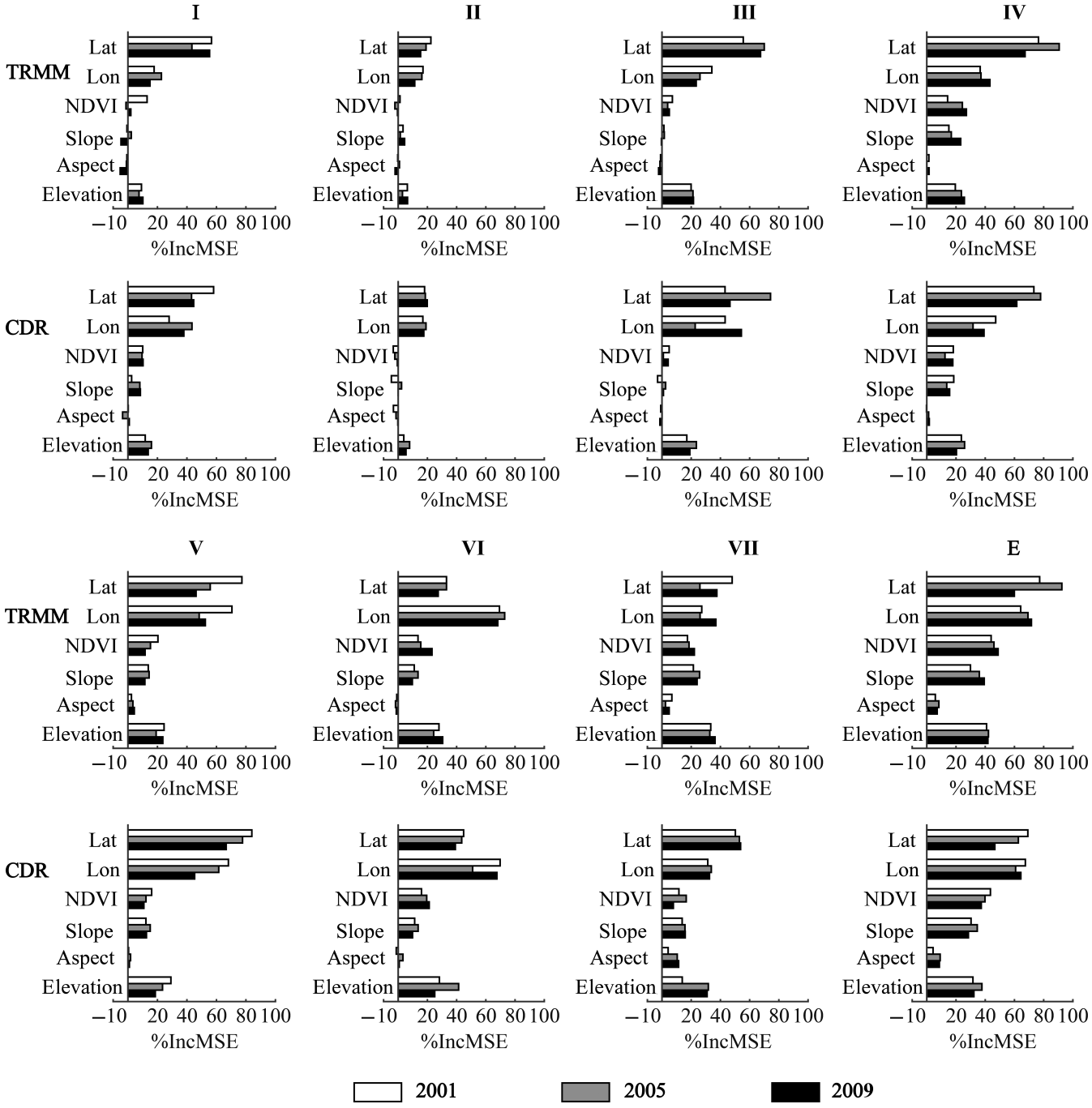

FIGURE 8 Relative importance of predictors for downscaled precipitation in individual precipitation subregions (I, II, III, IV, V, VI, and VII) and the entire basin (E) for the years of 2001, 2005, and 2009. The value of IncMSE (i.e., increase in MSE) is equivalent to the mean decrease accuracy of the random forest classification. The greater the values are, the higher the importance of the corresponding variables

product produce an east–west separation, while the CDR product yields more of a north–south separation (Figures 3a–c). This difference can be ascribed to the differences in the capabilities of capturing precipitation extremes among the three data sources (Chen *et al.,* 2018). In precipitation subregions IV–VII with high amounts and variability of precipitation, the CDR product has lower precipitation variation than that found in ground-based observations and the TRMM product (Figures 3e and 4, Table 1). Ground-based observations and the TRMM product capture the longitudinal gradient of precipitation extremes and produce an east–west separation, while the CDR product captures the latitudinal gradient of non-extreme precipitation variability and yields more of a north–south separation. The three precipitation data sources represent different aspects of regional precipitation and its variability. As a result, the REOF results differ among the different sources.

Because of the discontinuities in both time and space in the ground-based observational precipitation records, precipitation zoning was conducted using an iterative REOF analysis that combined data sets on ground- and satellite-based precipitation in this study. The resultant zoning based on the

combined data set was superior to that only based on ground-based precipitation. It succeeded in separating the Khorat Upland subregion from the Mekong Lowland subregion. The spatial patterns of annual precipitation in the two subregions were obviously different (Figure 4). Therefore, the iterative REOF analysis that combined ground- and satellite-based precipitation is more suitable for regions that have a scarce amount of ground observational data.

However, the relatively coarse spatial resolutions of the TRMM and CDR products might blur the boundaries of the resultant subregions. With the launch and operation of fine-resolution satellite precipitation platforms, such as the Global Precipitation Measurement (GPM) mission, the full integration of ground- and satellite-based precipitation for global or regional precipitation zoning is promising.

4.2 | Performance of the zoning-based downscaling approach

The validation of downscaled precipitation by comparing original satellite-based products and ground-based observations consistently showed that the Zrf outperformed the Drf. Compared to the Drf, the Zrf produced smaller precipitation estimation deviations and better statistical evaluation indices (Figures 4 and 5, 7 and Tables 1 and 2). This approach led to statistically significant reductions in RMSE and MAE, with a relative difference of 5–17% compared to the original satellite-based precipitation data and 1–5% compared to the ground observations. The RF regression model further shows that the relative importance is different for the entire basin and individual subregions (Figure 8). Across the north–south-oriented Lancang–Mekong River basin, geographical location (variables of Lat, Lon, and elevation) is the key determinant in the statistical modelling for most of the precipitation subregions, while the NDVI is useful in some instances (Figure 8). However, NDVI is second in importance only to the variables of Lat and Lon and is equally or more important than the variables of elevation and slope at the entire basin scale. This pattern reflects the differences in the dominant factors of precipitation distributions at different spatial scales. This finding is consistent with the results of previous studies (Grist *et al.,* 1997; Foody, 2003; Chen *et al.,* 2014; Xu *et al.,* 2015), which highlighted the spatial inconsistencies of the relationships between precipitation and geospatial variables. Grist *et al.* (1997) reported that the contribution of the NDVI to precipitation estimation is inconsistent at the regional scale (e.g., the semi-arid region of southern Botswana) and local scale (e.g., four subareas in southern Botswana, which have diverse soil conditions). Foody (2003) pointed out that the NDVI–precipitation relationship is spatially non-stationary in the region over North Africa and the Middle East, and a global regression analysis may omit local details that can be significant. Chen *et al.* (2014) found that an NDVI–precipitation relationship obtained at the local level cannot apply to other local or entire regions, and a global regression model at a vast regional scale can also become unsuitable at the local scale. In addition, Xu *et al.* (2015) observed a non-stationarity of the NDVI–precipitation or topography–precipitation relationships on the eastern Tibetan Plateau and in the Tianshan Mountains of western China. Thus, homogeneous subregions with similar dominant controlling factors can effectively remove noise variables and improve the accuracy of precipitation estimation.

The Zrf led to the less consistent improvement of the statistical evaluation indices among the examined precipitation subregions. Moreover, marked differences in the relative importance exist among the individual subregions. This result is supported by the results of Chen *et al.* (2017), who observed large spatial differences in the correlation between precipitation and geospatial variables in the upper Lancang–Mekong River basin. Regional differences in the relative importance of precipitation predictor variables imply disparities in the controlling factors of precipitation patterns among different precipitation subregions (Figure 8). This finding is consistent with the results of Jia *et al.* (2011) and Verlinde (2011), who reported the inconsistent effects of elevation on precipitation distributions between the Qaidam basin of China and the Pangani basin of Tanzania. In addition, a recent study suggested that the roles of geospatial predictors in precipitation estimation vary among different climate zones across Mainland China (Shi *et al.,* 2015). It is worth noting that the RF regression models in subregion II provide the lower variable importance of each predictor variable (Figure 8). This result reflects the fact that the selected predictor variables were unsuitable for predicting precipitation distributions in this subregion, which could be attributed to the unique terrain and related climate regimes characterized by extreme elevation and climate gradients from the valley bottom to the mountain ridge (Wu *et al.,* 2013). A substantial elevation decrease of approximately 3,000 m between the high mountains and deep valleys in subregion II occurs within a horizontal range of approximately 20 to 60 km (Figure 1). In addition, a smaller sample size in this subregion is an important factor that affects the stability and prediction accuracy of the RF regression model (Moreno *et al.,* 2010). Previous studies have also suggested that the correlation between the TRMM 3B43 data and ground observational precipitation at a monthly scale in the middle region of the Lancang River basin is very poor (Zeng *et al.,* 2012a, 2012b).

The performance of precipitation downscaling is also affected by the precipitation products used. Precipitation downscaling based on the CDR products outperforms that based on the TRMM products in subregions II, IV, VI, and VII (Figure 6), which is related to the reduced spatial variability in CDR precipitation in these subregions compared to that for TRMM precipitation (Figure 4). This result is supported by the results of Jing *et al.* (2016), who found that the

MAEs between the original and downscaled TRMM data tend to be higher over the southern part of the Tibetan Plateau, where a high precipitation regime and uneven spatial distribution exist. In addition, the performance of the Zrf varies with the precipitation product. Following the implementation of the Zrf, large improvements in the accuracy of precipitation estimation occur in subregions I and II via the CDR products compared to the TRMM products (Figure 6). Thus, it can be inferred that the performance of the Zrf is data-specific in some cases. This data-specific performance was also reported by Zhu *et al*. (2016), who found that the estimated values of extreme precipitation varied with the precipitation product, and extreme precipitation was often underestimated for the CDR products and uncertainly underestimated and overestimated for the TRMM products.

One important advantage of the RF regression model method is its insensitivity and robustness to outliers (Breiman, 2001; Liaw and Wiener, 2002). The effects are, however, twofold (i.e., the overestimation of lower values and underestimation of higher values). Across the entire Lancang–Mekong River basin, the amount of annual precipitation is extremely imbalanced among different regions, varying from approximately 300 to 5,000 mm. This extreme imbalance in the observed precipitation samples leads to the omission of samples with extremely high and low precipitation values when fitting the RF regression model (Del Río *et al.,* 2014). Therefore, lower precipitation values were obviously overestimated, and higher precipitation values were obviously underestimated (Figures 5 and 7). Several previous studies on the spatial downscaling of precipitation (Shi *et al.,* 2015; Jing *et al.,* 2016) and land surface temperature (Hutengs and Vohland, 2016) also indicated high and low-value deviations in their results via the RF regression models. In this study, the Zrf provided a better performance in removing the negative effects of the RF regression model than the Drf (Figures 5 and 7).

5 | CONCLUSIONS

Coarse spatial resolutions of approximately 0.1°–5° limit the popularization and application of remote sensing precipitation products. To generate high-spatial-resolution precipitation data, we proposed a zoning-based precipitation downscaling approach by integrating an iterative REOF analysis with random forest regression and applied this approach to downscale the 0.25°-resolution TRMM and CDR precipitation products to 1-km-resolution precipitation data across the entire Lancang–Mekong River basin.

By integrating the ground observational data with the TRMM and CDR precipitation products via an iterative REOF analysis, the Lancang–Mekong River basin can be broadly divided into seven precipitation subregions. The validations with the satellite- and ground-based observations consistently show that the Zrf can effectively improve the accuracy of the downscaled precipitation estimates, resulting in statistically significant reductions of approximately 1–17% in the RMSE and MAE at the entire basin scale and approximately 1–46% in the RMSE and MAE at the precipitation subregion scale. In addition, the Zrf can effectively offset the overestimation of lower precipitation values and underestimation of higher precipitation values associated with imbalanced sample sizes.

Although we used only TRMM and CDR precipitation products in this work, the Zrf proposed in this study can be extended to process other types of satellite-based precipitation data. The approach might be more promising for precipitation downscaling at continental and global scales due to the more complex spatial variabilities in precipitation and geospatial variables. However, more empirical research is still needed. In the future, a finer and more comprehensive zoning approach by coupling a large number of hydrometeorological–geomorphological factors with a higher spatial resolution should be considered to further improve the performance of precipitation downscaling models in the Lancang–Mekong River basin.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

REFERENCES

Ashouri, H., Hsu, K.L., Sorooshian, S., Braithwaite, D.K., Knapp, K.R., Cecil, L.D., Nelson, B.R., Prat, O. (2015) PERSIANN-CDR daily precipitation climate data record from multisatellite observations for hydrological and climate studies. *Bulletin of the American Meteorological Society*, 96(1), 69–83. https://doi.org/10.1175/BAMS-D-13-00068.1.

Atkinson, P.M. (2013) Downscaling in remote sensing. *International Journal of Applied Earth Observation and Geoinformation*, 22, 106–114. https://doi.org/10.1016/j.jag.2012.04.012.

Breiman, L. (2001) Random forests. *Machine Learning*, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324.

Brunetti, M.T., Melillo, M., Peruccacci, S., Ciabatta, L. and Brocca, L. (2018) How far are we from the use of satellite rainfall products in landslide forecasting? *Remote Sensing of Environment*, 210, 65–75. https://doi.org/10.1016/j.rse.2018.03.016.

Chen, A., Chen, D. and Azorin-Molina, C. (2018) Assessing reliability of precipitation data over the Mekong river basin: a comparison of ground-based, satellite, and reanalysis datasets. *International Journal of Climatology*, 14, 1–21. https://doi.org/10.1002/joc.5670.

Chen, F.R., Liu, Y., Liu, Q. and Li, X. (2014) Spatial downscaling of TRMM 3B43 precipitation considering spatial heterogeneity. *International Journal*

of Remote Sensing, 35(9), 3074–3093. https://doi.org/10.1080/01431161.2014.902550.

Chen, S.J., Li, L.J., Li, J.Y. and Liu, J.X. (2017) Analysis of the temporal and spatial variation characteristics of precipitation in the Lancang River basin over the past 55 years. *Journal of Geo-Information Science*, 19(3), 365–373 (in Chinese with English abstract). https://doi.org/10.3724/SP.J.1047.2017.00365.

Conroy, J.L. and Overpeck, J.T. (2011) Regionalization of present-day precipitation in the Greater Monsoon Region of Asia. *Journal of Climate*, 24(2), 4073–4095. https://doi.org/10.1175/2011JCLI4033.1.

Del Río, S., López, V., Benítez, M.J. and Herrera, F. (2014) On the use of MapReduce for imbalanced big data using Random Forest. *Information Sciences*, 285, 112–137. https://doi.org/10.1016/j.ins.2014.03.043.

Didan, K., Munoz, A.B., Solano, R. and Huete, A. (2015). MODIS vegetation index user's guide (MOD13 Series) version 3. https://lpdaac.usgs.gov/sites/default/files/public/product_documentation/mod13_user_guide.pdf.

Dos Santos, G.M.A.D.A., Dos Santos, A.R., Teixeira, L.J.Q., Saraiva, S.H., Freitas, D.F., Pereira, O.D.S., Ribeiro, C., Lorenzon, A.S., Eugenio, F.C., Neves, A.A., Queiroz, M.E.L.R., Scherer, R. (2016) GIS applied to agriclimatological zoning and agrotoxin residue monitoring in tomatoes: a case study in Espirito Santo state, Brazil. *Journal of Environmental Management*, 166, 429–439. https://doi.org/10.1016/j.jenvman.2015.10.040.

Fan, H. and He, D.M. (2015) Temperature and precipitation variability and its effects on streamflow in the upstream regions of the Lancang-Mekong and Nu-Salween Rivers. *Journal of Hydrometeorology*, 16(5), 2248–2263. https://doi.org/10.1175/JHM-D-14-0238.1.

Foody, G.M. (2003) Geographical weighting as a further refinement to regression modelling: an example focused on the NDVI-rainfall relationship. *Remote Sensing of Environment*, 88(3), 283–293. https://doi.org/10.1016/j.rse.2003.08.004.

Grist, J., Nicholson, S.E. and Mpolokang, A. (1997) On the use of NDVI for estimating rainfall fields in the Kalahari of Botswana. *Journal of Arid Environments*, 35(2), 195–214. https://doi.org/10.1006/jare.1996.0172.

Gupta, A. (2009) *Geology and Landforms of the Mekong Basin. The Mekong: Biophysical Environment of an International River Basin*. New York, NY: Academic Press, pp. 29–51. https://doi.org/10.1016/B978-0-12-374026-7.00003-6.

Harding, R.J., Weedon, G.P., Van Lanen, H.A.J. and Clark, D.B. (2014) The future for global water assessment. *Journal of Hydrology*, 518, 186–193. https://doi.org/10.1016/j.jhydrol.2014.05.014.

Hattab, T., Albouy, C., Lasram, F.B.R., Le Loc'h, F., Guilhaumon, F. and Leprieur, F. (2015) A biogeographical regionalization of coastal Mediterranean fishes. *Journal of Biogeography*, 42(7), 1336–1348. https://doi.org/10.1111/jbi.12505.

Haylock, M.R., Cawley, G.C., Harpham, C., Wilby, R.L. and Goodess, C.M. (2006) Downscaling heavy precipitation over the United Kingdom: a comparison of dynamical and statistical methods and their future scenarios. *International Journal of Climatology*, 26(10), 1397–1415. https://doi.org/10.1002/joc.1318.

He, Z.H., Yang, L., Tian, F.Q., Ni, G.H., Hou, A.Z. and Lu, H. (2017) Intercomparisons of rainfall estimates from TRMM and GPM multisatellite products over the Upper Mekong River basin. *Journal of Hydrometeorology*, 18(2), 413–430. https://doi.org/10.1175/JHM-D-16-0198.1.

Hou, A.Y., Kakar, R.K., Neeck, S., Azarbarzin, A.A., Kummerow, C.D., Kojima, M., Oki, R., Nakamura, K., Iguchi, T. (2014) The global precipitation measurement mission. *Bulletin of the American Meteorological Society*, 95(5), 701–722. https://doi.org/10.1175/BAMS-D-13-00164.1.

Hunink, J.E., Immerzeel, W.W. and Droogers, P. (2014) A high-resolution precipitation 2-step mapping procedure (HiP2P): development and application to a tropical mountainous area. *Remote Sensing of Environment*, 140, 179–188. https://doi.org/10.1016/j.rse.2013.08.036.

Hutengs, C. and Vohland, M. (2016) Downscaling land surface temperatures at regional scales with random forest regression. *Remote Sensing of Environment*, 178, 127–141. https://doi.org/10.1016/j.rse.2016.03.006.

Immerzeel, W.W., Rutten, M.M. and Droogers, P. (2009) Spatial downscaling of TRMM precipitation using vegetative response on the Iberian Peninsula. *Remote Sensing of Environment*, 113(2), 362–370. https://doi.org/10.1016/j.rse.2008.10.004.

IPCC. (2013) *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. New York, NY: Cambridge University Press, pp. 953–1136.

Jia, S.F., Zhu, W.B., Lu, A.F. and Yan, T.T. (2011) A statistical spatial downscaling algorithm of TRMM precipitation based on NDVI and DEM in the Qaidam Basin of China. *Remote Sensing of Environment*, 115(12), 3069–3079. https://doi.org/10.1016/j.rse.2011.06.009.

Jing, W.L., Yang, Y.P., Yue, X.F. and Zhao, X.D. (2016) A spatial downscaling algorithm for satellite-based precipitation over the Tibetan Plateau based on NDVI, DEM, and land surface temperature. *Remote Sensing*, 8(8), 655. https://doi.org/10.3390/rs8080655.

Kaiser, H.F. (1958) The varimax criterion for analytic rotation in factor-analysis. *Psychometrika*, 23(3), 187–200. https://doi.org/10.1007/BF02289233.

Karl, T.R., Koscielny, A.J. and Diaz, H.F. (1982) Potential errors in the application of principal component (eigenvector) analysis to geophysical data. *Journal of Applied Meteorology*, 21(8), 1183–1186. https://doi.org/10.1175/1520-0450(1982)021<1183:PEITAO>2.0.CO;2.

Kurina, F.G., Hang, S., Cordoba, M.A., Negro, G.J. and Balzarini, M.G. (2018) Enhancing edaphoclimatic zoning by adding multivariate spatial statistics to regional data. *Geoderma*, 310, 170–177. https://doi.org/10.1016/j.geoderma.2017.09.011.

Leinenkugel, P., Kuenzer, C., Oppelt, N. and Dech, S. (2013) Characterisation of land surface phenology and land cover based on moderate resolution satellite data in cloud prone areas: a novel product for the Mekong Basin. *Remote Sensing of Environment*, 136, 180–198. https://doi.org/10.1016/j.rse.2013.05.004.

Liaw, A. (2012). Package "randomForest." http://www.stat.berkeley.edu/users/breiman/RandomForests.

Liaw, A. and Wiener, M. (2002) Classification and regression by randomforest. *R News*, 2(3), 18–22.

MacRitchie, K., Huffman, G., and Bolvin, D. (2017). Readme document for the Tropical Rainfall Measurement Mission (TRMM) version 7. https://disc2.gesdisc.eosdis.nasa.gov/data/TRMM_L3/TRMM_3B43.7/doc/README.TRMM_V7.pdf.

Moreno, J.I.L., Latron, J. and Lehmann, A. (2010) Effects of sample and grid size on the accuracy and stability of regression-based snow interpolation methods. *Hydrological Processes*, 24(14), 1914–1928. https://doi.org/10.1002/hyp.7564.

MRC. (2005) *Overview of the Hydrology of the Mekong Basin*. Vientiane: Mekong River Commission.

Ochoa, A., Campozano, L., Sanchez, E., Gualan, R. and Samaniego, E. (2016) Evaluation of downscaled estimates of monthly temperature and precipitation for a Southern Ecuador case study. *International Journal of Climatology*, 36(3), 1244–1255. https://doi.org/10.1002/joc.4418.

Räsänen, T.A. (2014) *Hydrological Changes in the Mekong River Basin: The Effects of Climate Variability and Hydropower Development*. Helsinki, Finland: Aalto University.

Richman, M.B. (1986) Rotation of principal components. *International Journal of Climatology*, 6(3), 293–335. https://doi.org/10.1002/joc.3370060305.

Rundel, P.W. (2009) Vegetation in the Mekong Basin. In: *The Mekong: Biophysical Environment of an International River Basin*. New York, NY: Academic Press, pp. 143–160. https://doi.org/10.1016/B978-0-12-374026-7.00007-3.

Shi, Y.L., Song, L., Xia, Z., Lin, Y.R., Myneni, R.B., Choi, S., Wang, L., Ni, X., Lao, C., Yang, F. (2015) Mapping annual precipitation across mainland China in the period 2001-2010 from TRMM3B43 product using spatial downscaling approach. *Remote Sensing*, 7(5), 5849–5878. https://doi.org/10.3390/rs70505849.

Spracklen, D.V., Arnold, S.R. and Taylor, C.M. (2012) Observations of increased tropical rainfall preceded by air passage over forests. *Nature*, 489(7415), 282–285. https://doi.org/10.1038/nature11390.

Tan, M.L., Chua, V.P., Tan, K.C. and Brindha, K. (2018) Evaluation of TMPA 3B43 and NCEP-CFSR precipitation products in drought monitoring over Singapore. *International Journal of Remote Sensing*, 39(8), 2089–2104. https://doi.org/10.1080/01431161.2018.1425566.

Torsten, H., Kurt, H., Mark, A., Henric, W., and Achim, Z. (2017). Package "coin". http://coin.r-forge.r-project.org.

Ujeneza, E.L. and Abiodun, B.J. (2015) Drought regimes in Southern Africa and how well GCMs simulate them climate dynamics. *Bulletin of the American Meteorological Society*, 44(5–6), 1595–1609. https://doi.org/10.1007/s00382-014-2325-z.

Verlinde, J. (2011) *TRMM Rainfall Data Downscaling in the Pangani basin in Tanzania*. Delft, Netherlands: Delft University of Technology.

Wang, W., Lu, H., Yang, D.W., Sothea, K., Jiao, Y., Gao, B., Peng, X., Pang, Z. (2016) Modelling hydrologic processes in the Mekong River basin using a distributed model driven by satellite precipitation and rain gauge observations. *Plos One*, 11(3), e0152229. https://doi.org/10.1371/journal.pone.0152229.

Wang, W., Lu, H., Zhao, T., Jiang, L. and Shi, J. (2017) Evaluation and comparison of daily rainfall from latest GPM and TRMM products over the Mekong River basin. *IEEE Journal of Selected Topics in Applied Earth Observations & Remote Sensing*, 99, 1–10. https://doi.org/10.1109/JSTARS.2017.2672786.

White, D., Richman, M. and Yarnal, B. (1991) Climate regionalization and rotation of principal components. *International Journal of Climatology*, 11(1), 1–25. https://doi.org/10.1002/joc.3370110102.

Wu, F.F., Wang, X., Cai, Y.P. and Li, C.H. (2013) Spatiotemporal analysis of precipitation trends under climate change in the upper reach of Mekong River basin. *Quaternary International*, 392, 137–146. https://doi.org/10.1016/j.quaint.2013.05.049.

Xu, S., Wu, C., Wang, L., Gonsamo, A., Shen, Y. and Niu, Z. (2015) A new satellite-based monthly precipitation downscaling algorithm with non-stationary relationship between precipitation and land surface characteristics. *Remote Sensing of Environment*, 162, 119–140. https://doi.org/10.1016/j.rse.2015.02.024.

Zeng, H.W., Li, L.J., Hu, J.M., Liang, L.Q., Li, J.Y., Li, B. and Zhang, K. (2012b) Accuracy validation of TRMM multisatellite precipitation analysis daily precipitation products in the Lancang River basin of China. *Theoretical and Applied Climatology*, 112(3–4), 389–401. https://doi.org/10.1007/s00704-012-0733-8.

Zeng, H.W., Li, L.J. and Li, J.Y. (2012a) The evaluation of TRMM multisatellite precipitation analysis (TMPA) in drought monitoring in the Lancang River basin. *Journal of Geographical Sciences*, 22(2), 273–282. https://doi.org/10.1007/s11442-012-0926-1.

Zhang, J.H., Feng, Z.M., Jiang, L.G. and Yang, Y.Z. (2015) Analysis of the correlation between NDVI and climate factors in the Lancang river basin. *Journal of Natural Resources*, 30(9), 1425–1435 (in Chinese with English abstract). https://doi.org/10.11849/zrzyxb.2015.09.001.

Zhao, N., Yue, T.X., Chen, C.F., Zhao, M.W. and Fan, Z.M. (2018) An improved statistical downscaling scheme of tropical rainfall measuring Mission precipitation in the Heihe River basin, China. *International Journal of Climatology*, 38(8), 3309–3322. https://doi.org/10.1002/joc.5502.

Zhu, Q., Xuan, W.D., Liu, L. and Xu, Y.P. (2016) Evaluation and hydrological application of precipitation estimates derived from PERSIANN-CDR, TRMM 3B42V7, and NCEP-CFSR over humid regions in China. *Hydrological Processes*, 30(17), 3061–3083. https://doi.org/10.1002/hyp.10846.

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